NAI-WEN CHI

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RESEARCH INTERESTS

Drug delivery and biomaterials; lipid and polymeric nanoparticles; targeted nucleic-acid delivery; hydrogels and tissue repair; translational biomedical research.

EDUCATION

M.S., Biomedical Engineering | National Tsing Hua University, Taiwan Feb 2015 - Jan 2017

GPA: 4.05/4.30 | Thesis research: nitric-oxide (NO) delivery for osteoporosis therapy

B.S., Agronomy | National Chung Hsing University, Taiwan Sep 2008 - Jun 2012

GPA: 3.25/4.00

RESEARCH AND TRANSLATIONAL DEVELOPMENT EXPERIENCE

R&D and Regulatory Scientist | Coastar Therapeutics, California Dec 2025 - Present

- Identified CMC and nonclinical data gaps against FDA guidance for early-stage genetic-medicine programs.
- Performed LNP purification and concentration using dialysis and centrifugal ultrafiltration, followed by particle-size characterization using DLS.
- Proposed nonclinical models, experimental controls, and safety readouts for team evaluation.
- Led preparation of an FDA INTERACT briefing package in 2026, developing the document structure, coordinating contributions, and reviewing and refining technical content; consolidated meeting feedback and drafted follow-up responses.

Research Assistant | Academia Sinica, Taiwan | Dr. Yu-Jung Lin Apr 2025 - Nov 2025

- Prepared and characterized polymeric materials for nanoparticle-based drug delivery using analytical data from NMR, GPC, and MALDI-TOF.
- Prepared drug-loaded polymeric nanoparticles and compared preparation and purification conditions (dialysis versus ultrafiltration); measured particle size and zeta potential by DLS and drug loading and encapsulation efficiency by HPLC.
- Conducted in vitro drug-release studies and examined sink conditions in the release-study design; compared fresh and freeze-dried nanoparticles for size stability.
- Assessed nanoparticle cytotoxicity using CCK-8 assays, prepared liposomes, and measured the gelation time of alginate-based hydrogels prepared according to established protocols.

R&D Engineer | BioGend Therapeutics, Taiwan Jan 2018 - Mar 2021

- Developed tyramine-alginate hydrogel materials for cartilage repair and contributed to a study published in 2022.
- Supported PLGA/TCP scaffold preparation, mechanical testing, mini-pig studies, and usability testing for a cartilage repair implant.
- Contributed to the development of RevoCart, which received TFDA registration in 2020, through technical documentation and work with the development team.

Master's Research: Nitric Oxide Delivery for Osteoporosis | National Tsing Hua University 2015 - 2017

- Contributed to formulation design of temperature-responsive, phase-change carriers for sustained nitric oxide (NO) delivery, and measured NO release kinetics.
- Performed bone-related assays using MC3T3-E1 cells, including WST-1 viability, ALP activity, and Alizarin Red S staining.
- Evaluated treatment effects in an ovariectomized rat model using H&E histology, serum ALP and osteocalcin, and bone-density (CT-number) results provided by a collaborating facility.
- Co-authored a 2018 Advanced Materials paper and was named co-inventor on a related patent family.

R&D Engineer | Exactech Taiwan Ltd., Taiwan May - Dec 2017

- Supported research and clinical-development activities for Class III medical devices.

PEER-REVIEWED PUBLICATIONS

1. Lin YJ, Chen CC, **Chi NW**, Nguyen T, Lu HY, Nguyen D, Lai PL, Sung HW. In Situ Self-Assembling Micellar Depots that Can Actively Trap and Passively Release NO with Long-Lasting Activity to Reverse Osteoporosis. Advanced Materials. 2018;30(22):1705605. DOI: 10.1002/adma.201705605

2. Wong CC, Lu CX, Cho EC, Lee PW, **Chi NW**, Lin PY, Jheng PR, Chen HL, Mansel BW, Chen YM, Chen CH, Chuang EY. Calcium peroxide aids tyramine-alginate gel to crosslink with tyrosinase for efficient cartilage repair. International Journal of Biological Macromolecules. 2022;208:299-313. DOI: 10.1016/j.ijbiomac.2022.03.044

PATENTS AND PUBLISHED APPLICATIONS | CO-INVENTOR

Temperature-responsive composition (National Tsing Hua University): Taiwan invention patent TWI606829B (granted 2017; later lapsed); U.S. published application US20180161276A1 (abandoned).

Cartilage filling system and surgical instrument set (BioGend): Taiwan utility model TWM594960U; China utility model CN212730081U; U.S. published application US20210252227A1 (abandoned).

Cartilage volumetric device and surgical instrument set (BioGend): Taiwan utility model TWM594437U; China utility model CN212729944U.

Dry-type heating device (BioGend): Taiwan utility model TWM590460U; China utility model CN211454393U.

ADDITIONAL REGULATORY AND PROFESSIONAL EXPERIENCE

Founder & Project Manager | Wen Fu AI Co., Taiwan Jul 2023 - Mar 2024

- Managed technology projects, schedules, project scope, and client communication.

Senior Regulatory Affairs Specialist | Winnoz Technology, Taiwan May 2022 - Jul 2023

- Prepared TFDA medical-device applications and amendments, EU IVDR and U.S. FDA IVD documentation; supported ISO 13485 and QMS audit readiness.

Regulatory Affairs Specialist / Project Manager | BioGend Therapeutics, Taiwan Apr 2021 - May 2022

- Managed TFDA Class I-III application activities, preclinical evidence coordination, design controls, and international regulatory documentation.

SELECTED RESEARCH METHODS

Materials and delivery: polymer synthesis and modification, polymeric nanoparticles, liposomes, alginate hydrogels, scaffold preparation; purification by dialysis, centrifugal ultrafiltration, and freeze-drying.

Characterization: DLS, zeta potential, HPLC drug loading and encapsulation, NO release kinetics, in vitro release testing, gelation time, mechanical testing; sample preparation for NMR, GPC, and MALDI-TOF.

Biological studies: MC3T3-E1 osteoblast culture, CCK-8/WST-1 viability, ALP activity, Alizarin Red S; ovariectomized rat model (H&E histology, serum biomarkers, micro-CT data interpretation); mini-pig study support.

ADDITIONAL INFORMATION

Languages: Mandarin Chinese (native); English (working proficiency).